DATA ANALYSIS PYTHON PROJECT - BLINKIT ANALYSIS

Import libraries

In [1]:
```
import pandas as pd
import seaborn as sns
import numpy as np
import matplotlib.pyplot as plt
import plotly.express as px
```

Import Raw Data

In [2]:
```
df = pd.read_csv("E:/Python/Project/Blinkit Data/blinkit_data.csv")
```

Sample Data

In [3]:
```
df.head(10)
```

Out[3]:

	Item Fat Content	Item Identifier	Item Type	Outlet Establishment Year	Outlet Identifier	Outlet Location Type	Outlet Size	Outlet Type	\
0	Regular	FDX32	Fruits and Vegetables	2012	OUT049	Tier 1	Medium	Supermarket Type1	0
1	Low Fat	NCB42	Health and Hygiene	2022	OUT018	Tier 3	Medium	Supermarket Type2	0
2	Regular	FDR28	Frozen Foods	2010	OUT046	Tier 1	Small	Supermarket Type1	0
3	Regular	FDL50	Canned	2000	OUT013	Tier 3	High	Supermarket Type1	0
4	Low Fat	DRI25	Soft Drinks	2015	OUT045	Tier 2	Small	Supermarket Type1	0
5	low fat	FDS52	Frozen Foods	2020	OUT017	Tier 2	Small	Supermarket Type1	0
6	Low Fat	NCU05	Health and Hygiene	2011	OUT010	Tier 3	Small	Grocery Store	0
7	Low Fat	NCD30	Household	2015	OUT045	Tier 2	Small	Supermarket Type1	0
8	Low Fat	FDW20	Fruits and Vegetables	2000	OUT013	Tier 3	High	Supermarket Type1	0
9	Low Fat	FDX25	Canned	1998	OUT027	Tier 3	Medium	Supermarket Type3	0

In [4]:
```
df.tail(10)
```

Out[4]:

	Item Fat Content	Item Identifier	Item Type	Outlet Establishment Year	Outlet Identifier	Outlet Location Type	Outlet Size	Outlet Type
8513	Regular	DRY23	Soft Drinks	1998	OUT027	Tier 3	Medium	Supermarket Type3
8514	low fat	FDA11	Baking Goods	1998	OUT027	Tier 3	Medium	Supermarket Type3
8515	low fat	FDK38	Canned	1998	OUT027	Tier 3	Medium	Supermarket Type3
8516	low fat	FDO38	Canned	1998	OUT027	Tier 3	Medium	Supermarket Type3
8517	low fat	FDG32	Fruits and Vegetables	1998	OUT027	Tier 3	Medium	Supermarket Type3
8518	low fat	NCT53	Health and Hygiene	1998	OUT027	Tier 3	Medium	Supermarket Type3
8519	low fat	FDN09	Snack Foods	1998	OUT027	Tier 3	Medium	Supermarket Type3
8520	low fat	DRE13	Soft Drinks	1998	OUT027	Tier 3	Medium	Supermarket Type3
8521	reg	FDT50	Dairy	1998	OUT027	Tier 3	Medium	Supermarket Type3
8522	reg	FDM58	Snack Foods	1998	OUT027	Tier 3	Medium	Supermarket Type3

Size of Data

In [5]:
```
print("Size of Data:",df.shape)
```

```
Size of Data: (8523, 12)
```

Field info

In [6]:
```
df.columns
```

Out[6]:
```
Index(['Item Fat Content', 'Item Identifier', 'Item Type',
       'Outlet Establishment Year', 'Outlet Identifier',
       'Outlet Location Type', 'Outlet Size', 'Outlet Type', 'Item Visibility',
       'Item Weight', 'Sales', 'Rating'],
      dtype='object')
```

Data Types

In [7]:
```
df.dtypes
```

Out[7]:
```
Item Fat Content              object
Item Identifier               object
Item Type                     object
Outlet Establishment Year      int64
Outlet Identifier             object
Outlet Location Type          object
Outlet Size                   object
Outlet Type                   object
Item Visibility              float64
Item Weight                  float64
Sales                        float64
Rating                       float64
dtype: object
```

Data Cleaning

In [8]:
```
print(df['Item Fat Content'].unique())
```

```
['Regular' 'Low Fat' 'low fat' 'LF' 'reg']
```

In [9]:
```
df['Item Fat Content'] = df['Item Fat Content'].replace({'LF':'Low Fat',
                                                         'low fat':'Low Fat',
                                                         'reg':'Regular'
                                                        })
```

In [10]:
```
print(df['Item Fat Content'].unique())
```

```
['Regular' 'Low Fat']
```

BUSINESS REQUIREMENTS

KPI's REQUIREMENTS

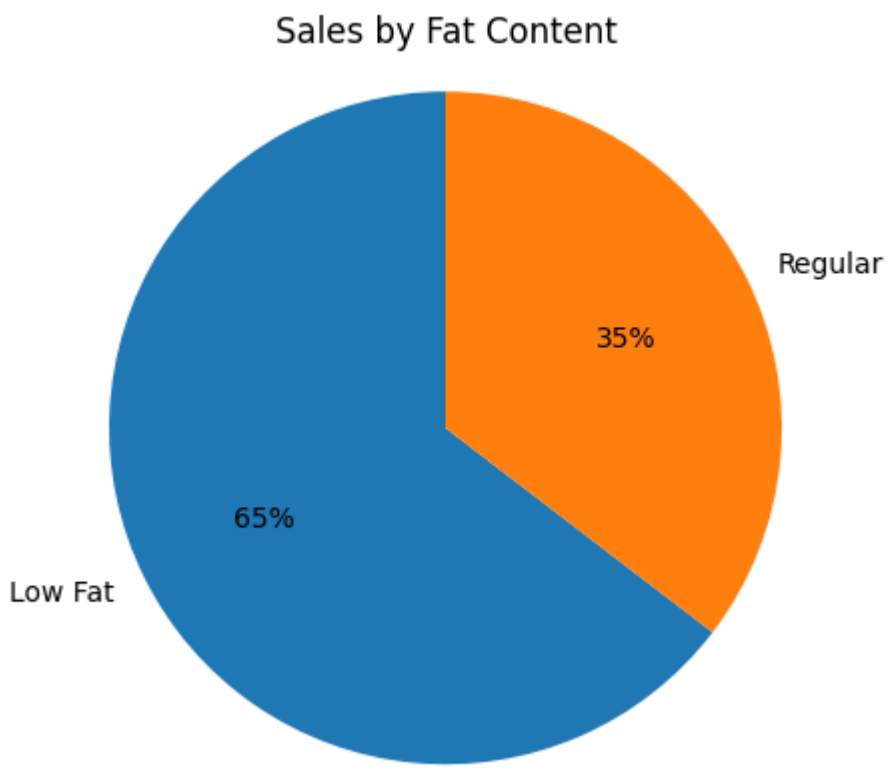
In [11]:
```
#Total Sales
total_sales = df['Sales'].sum()

#Average Sales
avg_sales=df['Sales'].mean()

#No of Items Sold
no_of_items_sold = df['Sales'].count()

#Average Ratings
avg_ratings = df['Rating'].mean()

#Display

print(f"Total Sales: ${total_sales:,.0f}")
print(f"Avarge Sales: ${avg_sales:,.1f}")
print(f"No of Items Sales: {no_of_items_sold:,.0f}")
print(f"Avarage Ratings: {avg_ratings:,.0f}")
```

```
Total Sales: $1,201,681
Avarge Sales: $141.0
No of Items Sales: 8,523
Avarage Ratings: 4
```

CHARTS REQUIREMENTS

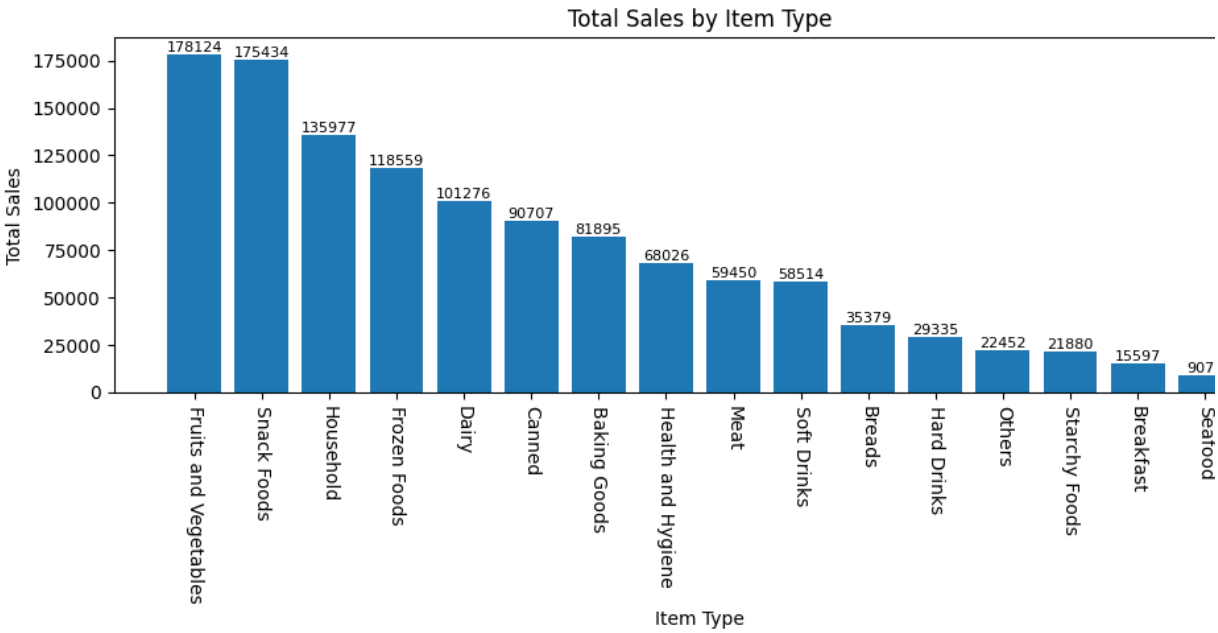
Total Sales by Fat Content

In [12]:
```
sales_by_fat = df.groupby('Item Fat Content')['Sales'].sum()
plt.pie(sales_by_fat, labels= sales_by_fat.index,
                            autopct = '%.0f%%',
                            startangle = 90)
plt.title('Sales by Fat Content')
plt.axis('equal')
plt.show()
```



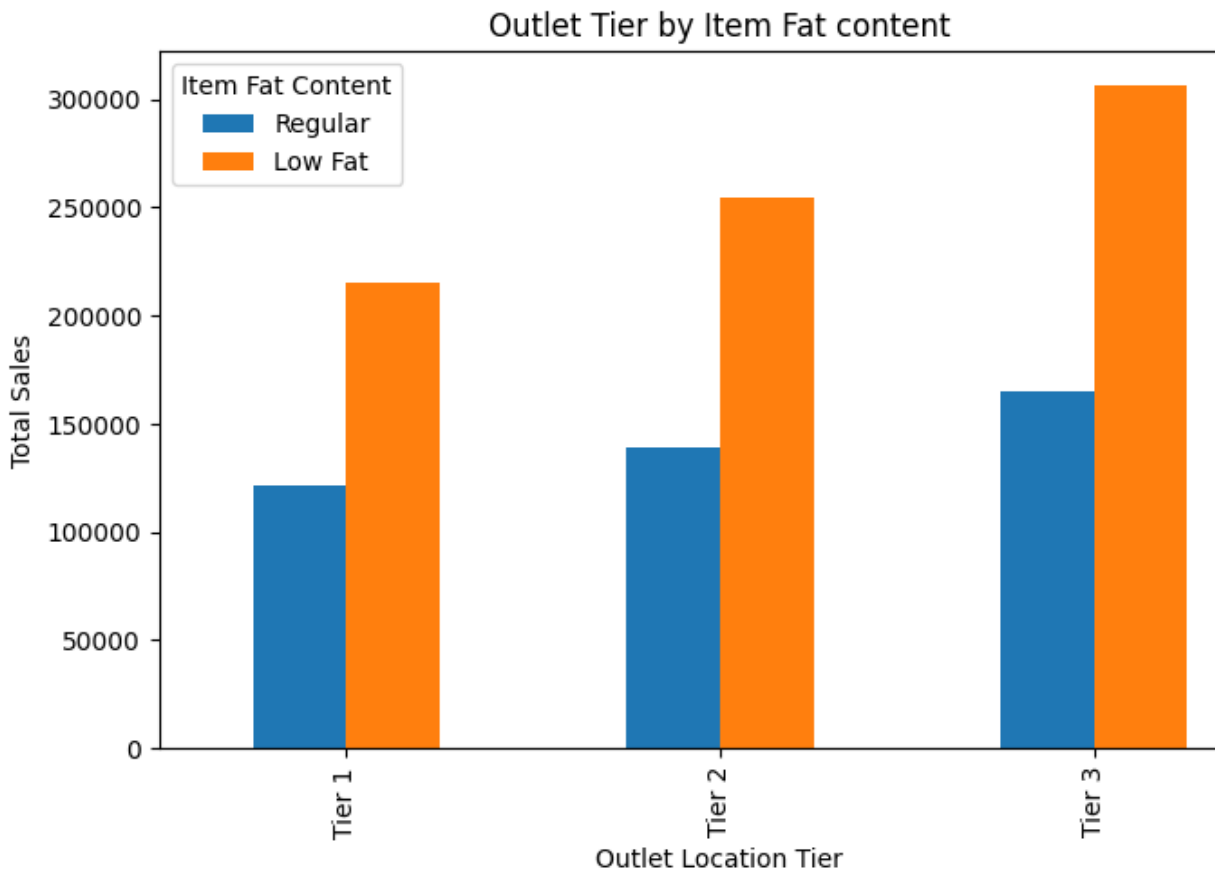
Total Sales by Fat Content

In [13]:
```
sales_by_type = df.groupby('Item Type')['Sales'].sum().sort_values(ascending=Fal

plt.figure(figsize=(10,5))
bars = plt.bar(sales_by_type.index, sales_by_type.values)

plt.xticks(rotation=-90)
plt.xlabel('Item Type')
plt.ylabel('Total Sales')
plt.title('Total Sales by Item Type')

for bar in bars:
    plt.text(bar.get_x() + bar.get_width() /2,bar.get_height(),
             f'{bar.get_height():.0f}', ha='center',va='bottom',fontsize=8)
plt.tight_layout()
plt.show()
```



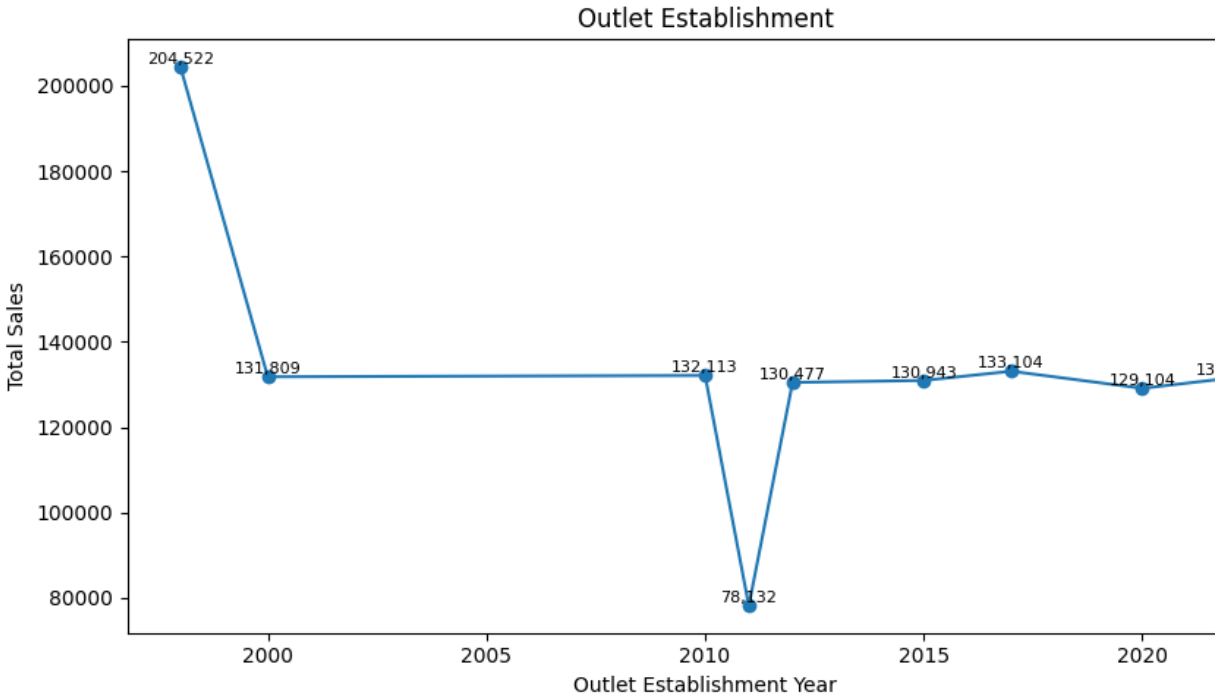
Fat content by Outlet forTotal Sales

In [14]:
```
grouped = df.groupby(['Outlet Location Type','Item Fat Content'])['Sales'].sum()
grouped = grouped[['Regular','Low Fat']]

ax = grouped.plot(kind='bar', figsize=(8,5), title='Outlet Tier by Item Fat cont
plt.xlabel('Outlet Location Tier')
plt.ylabel('Total Sales')
plt.legend(title='Item Fat Content')
plt.show()
```



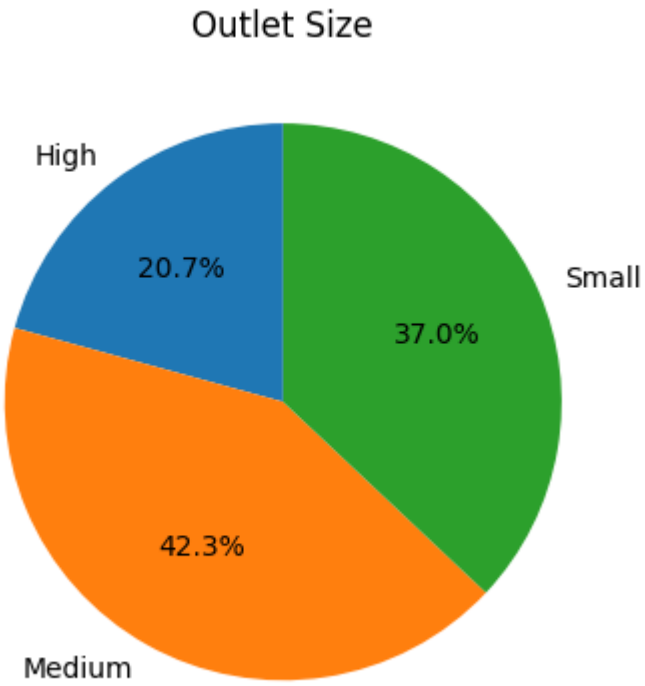
Total Sales by Outler Establishment

In [15]:
```
sales_by_year = df.groupby('Outlet Establishment Year')['Sales'].sum().sort_inde

plt.figure(figsize=(9,5))
plt.plot(sales_by_year.index, sales_by_year.values,marker ='o', linestyle='-')

plt.xlabel('Outlet Establishment Year')
plt.ylabel('Total Sales')
plt.title('Outlet Establishment')

for x,y in zip(sales_by_year.index, sales_by_year.values):
    plt.text(x,y,f'{y:,.0f}', ha='center', va='bottom',fontsize=8)

plt.tight_layout()
plt.show()
```



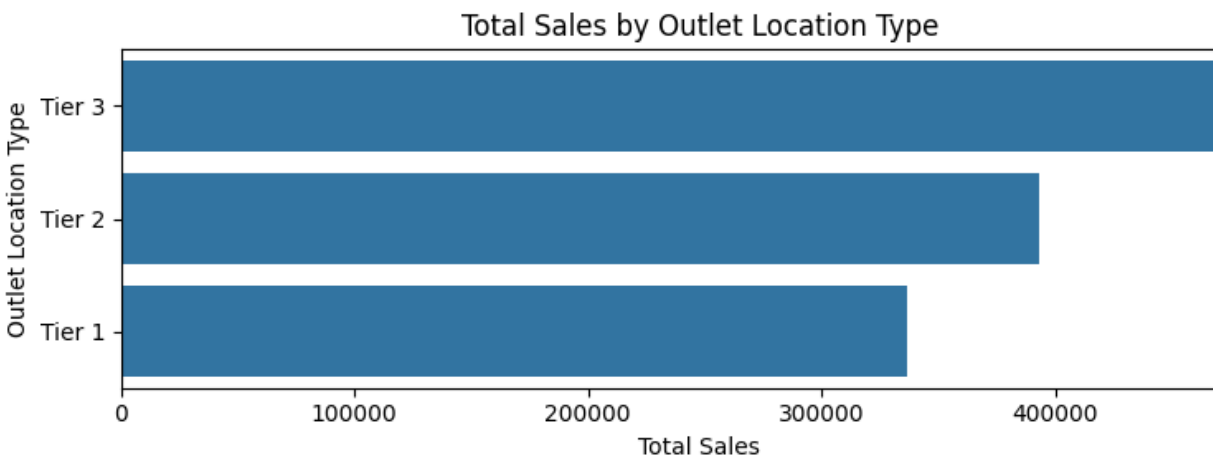
Sales by Outlet Size

In [16]:
```
sales_by_size = df.groupby('Outlet Size')['Sales'].sum()

plt.figure(figsize=(4,4))
plt.pie(sales_by_size, labels=sales_by_size.index, autopct='%1.1f%%',startangle=
plt.title('Outlet Size')
plt.tight_layout()
plt.show()
```



Sales by Outlet Location

In [18]:
```
sales_by_location = df.groupby('Outlet Location Type')['Sales'].sum().reset_inde
sales_by_location = sales_by_location.sort_values('Sales',ascending=False)

plt.figure(figsize=(8,3))
ax = sns.barplot(x='Sales',y='Outlet Location Type', data=sales_by_location)

plt.title('Total Sales by Outlet Location Type')
plt.xlabel('Total Sales')
plt.ylabel('Outlet Location Type')

plt.tight_layout()
plt.show()
```



In [ ]: